

Resistance and Resistivity

NB: resistivity is notated by the symbol ρ , and has units Ωm .

What affects resistance?

- The temperature of the wire
- The material of the wire
- The length of the wire $=: L$
- The cross-sectional area of the wire $=: A$

common resistivity values for different materials	
conductors	$10^{-8} \Omega m$
semiconductors	$10^{-5} \Omega m$ to $10^8 \Omega m$
insulators	$10^{16} \Omega m$

Resistance vs. Resistivity

The term 'resistance' refers only to a specific component. The term **resistivity** is used to describe the electrical property of a **material**.

Different components made from the same material may have different resistances due to their lengths/cross-sectional areas differing, but the **material** has a unique **resistivity**.

Here are some useful equations to know when dealing with resistances: $R \propto L$ and $R \propto \frac{1}{A}$.

An especially important equation to note is

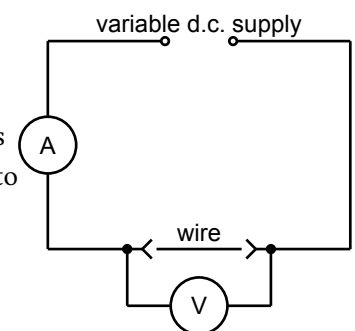
$$R = \frac{\rho L}{A}$$

where ρ is the constant of proportionality in the equation: the resistivity of some material at some temperature.

Resistivity varies with temperature the same way resistance varies with temperature: as T increases, so does ρ .

Determining ρ

Using this circuit, we can obtain values for the p.d. across different lengths of the wire. If the current in the wire is measured too, we can use $R = \frac{V}{I}$ to calculate the resistance for each length.



As $R \propto L$, a graph of R against L is a straight line through the origin. By considering the general equation for a straight line $y = mx + c$ and the equation $R = \frac{\rho L}{A}$ we see that the gradient of this graph is $\frac{\rho}{A}$.

